

Intelligent Reflecting Surfaces-Aided Authentication Mechanism for IoT Devices

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Abstract—The proliferation of Internet of Things (IoT) devices necessitates robust authentication mechanisms to ensure security and privacy in resource-constrained environments. Existing authentication protocols face limitations, including vulnerability to physical attacks, high computational overhead, and scalability challenges. This article presents a novel intelligent reflecting surface (IRS)-aided authentication mechanism for IoT devices that leverages IRSs to enhance location-based authentication through controllable signal enhancement. Our approach employs IRSs to amplify received signal strength (RSS) variations in response to device mobility, enabling precise location change detection while integrating pseudo-identity verification and RSS filtering to mitigate denial-of-service (DoS) attacks. Comprehensive robustness analysis demonstrates exceptional performance under realistic deployment constraints: simulations with Rician channel models show 99.82%–99.99% authentication performance across commercial IRS implementations (1-bit to 3-bit phase shifters), with 4.6-dB signal enhancement over direct paths and minimal system overhead (<0.03%). The mechanism achieves high suitability for infrastructure-dense deployments, including smart buildings and industrial IoT, while providing comprehensive protection against major attack vectors with lower computational complexity compared to existing protocols.

Index Terms—Authentication, intelligent reflecting surfaces (IRSs), Internet of Things (IoT) devices, location proximity, mobility support, robustness analysis.

I. INTRODUCTION

WITH the increasing prevalence of Internet of Things (IoT) deployments, safeguarding against cybersecurity threats has emerged as a paramount priority to sustain its continued growth [1]. Authentication of IoT devices is crucial for enhancing IoT security, significantly reducing the vulnerabilities associated with security breaches and data compromise by ensuring that only authorized and trusted devices can access the network.

Authentication mechanisms rely on secure credentials, such as digital certificates, keys, or tokens, to establish trust between devices and the network. Before 2000, authentication in IoT systems mainly relied on usernames and passwords, as the

early development stages focused primarily on functionality and connectivity. Subsequently, authentication mechanisms were enhanced by introducing intrinsic device/user-dependent credentials, including physical characteristics of hardware, such as the physically unclonable function (PUF) [2], [3], [4] or true random number generators (TRNGs) [5], [6], and biometric technologies like fingerprints, irises, voice recognition, or DNA [7]. The diverse credentials have facilitated the emergence of two-factor authentication (2FA) [8], [9], which leverages factors from two distinct categories to strengthen the authentication process.

Over the past decade, driven by technological advancements, IoT authentication mechanisms have evolved by incorporating innovative technologies like machine learning (ML) or blockchain. In an ML-based authentication mechanism, ML algorithms such as convolutional neural networks (CNNs), recurrent neural networks (RNNs), and support vector machines (SVMs) are utilized to analyze a wide range of biometric [10], [11], [12] and behavioral characteristics [13], [14], [15], thereby evaluating the validity of user identities. These ML-based authentication mechanisms offer significant advantages in terms of enhanced accuracy and resilience against attacks. However, their effectiveness depends on the quality of the data and the complexity of the ML algorithms employed, both of which demand substantial resource requirements [16]. This can pose a significant challenge for resource-constrained IoT devices. Additionally, ML-based authentication mechanisms may raise privacy concerns and encounter adaptability challenges [13]. In blockchain-based authentication mechanisms [17], [18], [19], [20], [21], [22], IoT devices are treated as nodes within the blockchain. Information for each device is registered in the blockchain ledger, and various actions such as device registration, identity authentication, and integrity verification are recorded as transactions. When a device requests network access, only registered devices whose information can be retrieved from the blockchain ledger can be authenticated successfully. Blockchain technology has pioneered decentralized peer-to-peer authentication with resistance to attacks. However, it faces scalability challenges associated with energy consumption, storage requirements, and communication costs, as it can impose additional time delays and increase network congestion [23], [24].

Despite exhibiting strong performance, authentication mechanisms based on ML or blockchain have limited practicality due to their resource-intensive nature, making them challenging to implement in resource-constrained IoT

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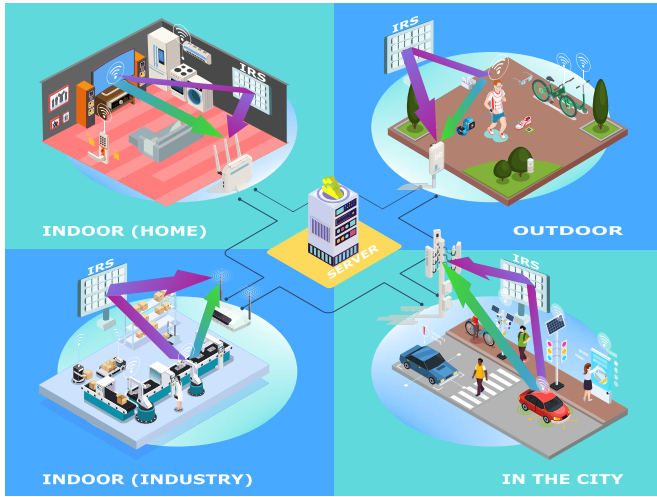


Fig. 1. Deployment of IRSs across residential, industrial, and urban IoT environments, highlighting its capabilities on signal improvement and enabling location-based authentication.

environments. In consideration of the IoT characteristics, there arises a demand for low-cost, resource-efficient, and highly secure authentication mechanisms. To address these concerns, we propose a novel IoT authentication mechanism based on intelligent reflecting surfaces (IRSs) in this article.

The IRS is a metasurface technology composed of a multitude of reconfigurable passive reflecting elements that can be individually adjusted through an intelligent control system to introduce phase shifts to the incident electromagnetic waves, thereby optimizing signal transmission performance [25], [26]. The IRS technology has been actively researched for 5G and beyond networks to address challenges related to signal propagation, coverage, and capacity in deployments [27]. The integration of IRSs into the IoT network enhances security and latency without significantly increasing resource consumption [28], [29]. With the ability to control and limit signal propagation, an IRS-aided system is more challenging for eavesdroppers to intercept communication [30]. The IRS does not introduce any latency into the communication system, and the IRS can even minimize delays in the system by optimizing communication resource allocation [31]. The IRS itself is passive and low-cost, and its integration does not necessitate any alterations to the existing system [27]. We take advantage of the fact that the IRS can enhance signal strength in controllable directions to build a location-based authentication mechanism. As shown in Fig. 1, in our proposed IRS-aided IoT systems model, an IRS is integrated into the IoT system to introduce an additional signal reflection path between devices and the gateway. This results in signal enhancement for wireless communication between devices at authentic locations and the gateway, while communication for devices at nonauthentic locations remains unaffected. Thus, we can identify illegitimate devices from nonauthentic locations by monitoring the received signal strength (RSS).

The main contributions of this article are as follows.

- 1) Incorporating advanced IRS technology into IoT systems for enhanced location-based identity authentication.

- 2) Protecting IoT device mobility by enabling authentication at multiple registration locations.
- 3) Integrating pseudo-identity verification and initial RSS verification to filter traffic before other authentication steps, significantly mitigating the risk of denial-of-service (DoS) attacks.

The rest of the article is organized as follows. The related works are discussed in Section II. The background knowledge about IRSs is briefed in Section III. The system model and assumptions are introduced in Section IV. Then, in Sections V and VI, we introduce the theoretical basis and the authentication protocol of the proposed IRS-based authentication mechanism, respectively. The security analysis is provided in Section VII. The performance of the proposed mechanism is discussed in Section VIII. The proposed mechanism's positioning is discussed in Section IX. In the end, we conclude the article in Section X.

II. RELATED WORKS

A. Studies on Location-Based Authentication Methods

Location-based IoT authentication methods use the physical location or proximity of devices as a factor for authentication. To verify a device's presence in a specific geographical area, the location-awareness technologies these methods rely on can be primarily categorized into two types.

1) *Direct Location Information*: Abdulrahman et al. [32] hash the location information obtained from the global positioning system (GPS) with the timestamp and preshared numbers to generate security tokens for authentication comparison. The authentication method proposed by Jin et al. [33] and Ambika [34] utilizes the hash value of the device's GPS coordinates to verify whether the device is attempting access within the designated area, all while protecting the confidentiality of the device's location information. Obtaining location information directly from GPS simplifies the authentication scheme, but also inherits the vulnerabilities associated with GPS, including susceptibility to GPS spoofing attacks. Additionally, most IoT devices are not equipped with GPS modules due to cost and complexity constraints.

2) *Indirect Location Information*: Chen et al. [35] combined channel state information (CSI) with smart contracts to design an authentication scheme for IoT devices to access Wi-Fi networks, in which the CSI information returned by relevant devices is used to confirm whether the requesting device is in a legal location. Aman et al. [8] combined the RSS indicator (RSSI) with the link quality indicator (LQI) for location verification, which is one of the authentication factors. Alqahtani et al. [36] introduced a method for authenticating new devices within a specific proximity to their own location. The distance to the new device is determined using the RSSI values collected from each available access point within the ad hoc network. Zhang et al. [37] utilized the RSS-variations resulting from relative horizontal displacement or rotation between the device and the access point for identity authentication. RSS/CSI-based approaches can be a lightweight and efficient choice for IoT scenarios. Since they exploit the inherent properties of wireless communications,

there are no additional costs associated with specialized hardware or tokens, making their implementation cost-effective. Likewise, their inherent properties allow for fast processing, resulting in low authentication latency. However, the traditional RSS/CSI-based methods require improvement in terms of accuracy and security. Wireless channels are dynamic and subject to rapid changes due to factors such as mobility, interference, and environmental conditions, making it difficult to maintain up-to-date authentication channel information. These methods are also vulnerable to spoofing attacks, where attackers can generate counterfeit RSS or CSI values to impersonate legitimate devices or generate counterfeit signals to disrupt legitimate authentication. Some works [8], [37] increase the computational complexity by combining multiple authentication factors (such as RSSI with PUF or CSI with smart contracts) to enhance security against spoofing attacks. However, this approach introduces significant computational overhead and resource requirements that may not be suitable for resource-constrained IoT devices. In contrast, our IRS-aided approach achieves similar security improvements through controllable signal enhancement without requiring additional computational complexity on the device side.

B. Studies on IRS-Assisted Authentication

Thanks to the IRS's capability to enhance signal strength in a specific direction, utilizing the IRS can perfectly inherit the advantages of the RSS method while enhancing accuracy and security. Several studies have demonstrated that integrating the IRS can enhance the resilience of physical-layer authentication against attacks by evaluating channel features such as channel impulse response (CIR) [38], [39], path loss [39], and RSS [40], [41]. Among these, only Shawky et al. [40] have proposed a specific protocol based on the elliptic curve digital signature algorithm (ECDSA); however, their analysis was limited to nonline-of-sight (NLoS) scenarios, considering only the reflection path in the received signal model.

Based on the above discussion, we present a comprehensive mechanism that fully incorporates the role of IRSs by accounting for both reflected and direct paths in the received signal modeling. Additionally, we introduce an efficient authentication protocol tailored for IRS-equipped systems.

III. BACKGROUND ON IRSs

An IRS is a technology used to enhance wireless communication systems, specifically improving signal quality and coverage. In this section, we provide fundamental background on three aspects of IRS: individual phase shift generation in IRSs and channel estimation techniques to facilitate comprehension of the subsequent content.

A. Phase Shift Generation

Comprising passive reflecting elements, the IRS manipulates and redirects radio waves by adjusting phase and amplitude to boost wireless network performance, including the increase in signal strength at specific locations, interference reduction, or coverage extension. Each reflecting element of the IRS is connected to a tunable chip capable of altering its load

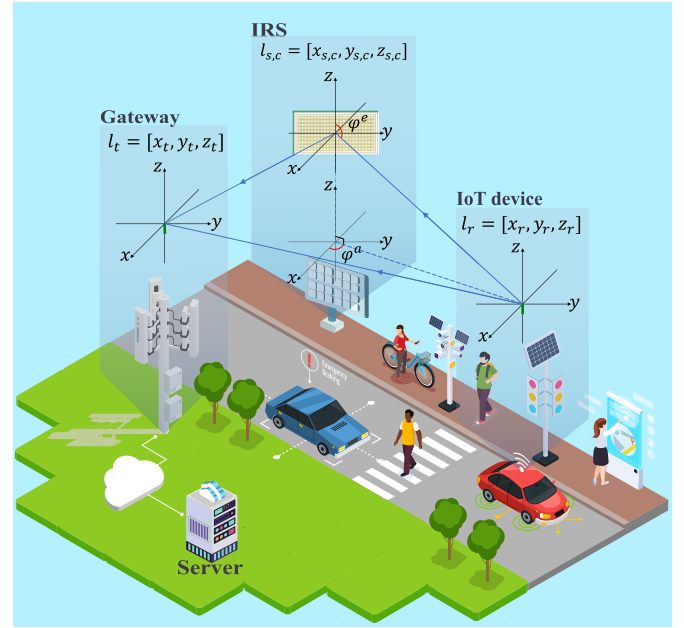


Fig. 2. System model showing device deployment and the spherical coordinate system.

impedance to produce a phase shift, achieved through either p-i-n diodes for a π phase shift difference or varactors for continuous phase shift, viz. bias voltage control. For simplicity, we consider that the IRS we use can provide continuous phase shifts.

B. Channel Estimation Techniques

The IRS typically uses preliminary channel estimation to determine how to configure the phase shifts of each reflecting element to enhance the signal in the specified user direction. Existing channel estimation methods can be categorized into four classes [46]: classical channel estimation [42], compressed sensing [43], [44], matrix factorization [44], and deep learning [43]. We provide the computational complexity for a single user's channel estimation with a fully passive IRS in Table I. It is worth noting that after the initial channel estimation, there is no computational complexity during the subsequent signal reflection process.

IV. SYSTEM MODEL AND ASSUMPTIONS

A. System Model

We consider a comprehensive system model encompassing multiple scenarios, as illustrated in Fig. 1, where we assume that the IRS diameter (D) is small, satisfying the condition ($D^2 < 1/2d\lambda_c$) [47], enabling the applicability of the far-field model. Due to limitations imposed by the signal transmission range, we divide the area into multiple subareas, each centered around a gateway unit. As depicted in Fig. 2, each subarea is equipped with a wireless gateway and an IRS designed for signal enhancement and device localization serving multiple IoT devices. In this configuration, IoT devices communicate with the server through the gateway in their respective area.

TABLE I
EXAMPLES OF COMPUTATIONAL COMPLEXITY FOR SEVERAL IRS CHANNEL ESTIMATION METHODS

Existing Works	Estimation Methods	Complexity ¹
Mishra et al. [42]	Least Square (LS) estimation	$\mathcal{O}(M^4 + M_t M^3)$
Mao et al. [43]	Double-Structured Orthogonal Matching Pursuit (DS-OMP) and Deep Learning (DL)	$\mathcal{O}(M_t(MN_C + L^2(L + N_C)) + \sum_{l=1}^{N_n} M_t M S_l C_{l-1} C_l)^2$
He et al. [44]	Bilinear Generalized Approximate Message (Bi-GAM) and Riemannian Gradient (RiGrad) algorithm	$IJO(MM_r T) + QO(rMT) + \mathcal{O}(M_t^2 T + M_t^3 + M_t MT)$ ³
Chen et al. [45]	Direction of Arrival (DOA) and maximum likelihood estimation	$\mathcal{O}(L^2 M_t)$

¹ M_t and M_r represent the number of transmitter and receiver antennas, respectively, M represent the number of reflecting elements of IRS, and L refer to the number of transmitter-to-IRS links.

² N_C is the number of time slots for CSI acquisition, N_n is the number of network layers, and S_l is the area of the convolution kernel of layer l , C_{l-1} and C_l are the number of input and output feature maps of layer l , respectively.

³ I and J are the maximum numbers of the outer and inner iterations of the BiG-AMP, Q is the maximum number of iterations of the RGrad, T is the sample size of coherence time of block fading channel, and r is rank value.

The gateway connects to the IRS controller via a wired connection to configure the IRS.

Simultaneously, we have depicted a 3-D Cartesian coordinate system for the locations of the gateway, the IRS, and an IoT device in Fig. 2. The antenna of the gateway is located at $\mathbf{l}_r = [x_r, y_r, z_r] \in \mathbb{R}^{1 \times 3}$. The uniform planar array (UPA) of the IRS is placed parallel to the yz -plane. The location of the IRS is $\mathbf{L}_s = [\mathbf{l}_{s,1}, \dots, \mathbf{l}_{s,m}, \dots, \mathbf{l}_{s,M}] \in \mathbb{R}^{M \times 3}$, and its center is located at $\mathbf{l}_{s,c} = [x_{s,c}, y_{s,c}, z_{s,c}] \in \mathbb{R}^{1 \times 3}$. The antenna of the IoT device is located at $\mathbf{l}_t = [x_t, y_t, z_t] \in \mathbb{R}^{1 \times 3}$.

1) *Communication Model*: We assume both the wireless gateways and the IoT devices have a single isotropic antenna, the IRS antennas are in a UPA, and the server communicates with only one IoT device in each session. Accordingly, our discussions are based on the single-input-single-output (SISO) communication model [48]. Signals are transmitted at power P , with a carrier frequency of f_c and a wavelength of λ_c .

2) *Communication Channel*: For the communication channels between the gateway and the IRS, we presume the presence of line-of-sight (LoS) components, leading us to consider Rician fading channels [48]

$$\mathbf{u} = \sqrt{\frac{\kappa}{1+\kappa}} \bar{\mathbf{u}} + \sqrt{\frac{1}{1+\kappa}} \tilde{\mathbf{u}} \quad (1)$$

where κ denotes the Rician fading factor and $\bar{\mathbf{u}}$ and $\tilde{\mathbf{u}} \sim \mathcal{CN}(0, \mathbf{I}_M)$ denote the LoS and NLoS components, respectively.

For the communication channels between the IRS and the IoT device, as well as the direct channel between the gateway and the IoT device, the selection between Rayleigh or Rician fading channels depends on factors such as the distance between the gateway and the IoT device, and the environmental conditions at their respective locations, as follows.

- 1) *Rician*: In the presence of LoS components, we utilize the Rician channel model as defined in (1).
- 2) *Rayleigh*: In the absence of LoS components, we adopt the Rayleigh channel model as the special case of (1) with $\kappa = 0$.
- 3) *Path Loss*: According to Friis [49], the ratio of the received power P_r to the transmitted power P_t is

$$\frac{P_r}{P_t} = \frac{A_r A_t}{d^2 \lambda_c^2} \quad (2)$$

where d represents the distance between the receiving and transmitting antennas, whereas A_r and A_t represent the effective areas of the receiving and transmitting antennas, respectively.

In our model, the effective area of the gateway and device antennas is $A_{\text{isso}} = \lambda_c^2 / (4\pi)$. Since they are equipped with isotropic antennas, the effective area of each element of IRSs is $A_e = \lambda_c^2 / (4\pi) G$, where G represents the antenna gain. In this article, let $\rho = (P_r/P_t)^{1/2} = (\lambda_c \sqrt{G}) / (4\pi d)$ to account for the path loss.

4) *Communication Channel Security Implementation*: The following layered security approach (encryption, authentication, integrity checking, and monitoring) ensures that the authentication infrastructure itself is protected. Periodic monitoring of authentication success rates and RSS measurement patterns enables early detection of potential attacks or system degradation.

- 1) *Server–Gateway Link*: TLS 1.3 with mutual certificates, AES-256-GCM encryption, session overhead ~ 50 ms amortized over hundreds of authentications.
- 2) *Gateway–IRS Link*: AES-128-GCM with preshared keys, HMAC-SHA256 authentication, control overhead $< 0.1\%$ (infrequent recalibration).

B. Assumptions

We make the following assumptions for the proposed authentication mechanism in accordance with the system model.

1) *Channel Security*: We assume the following regarding channel model:

- 1) The communication between the server and the gateway is considered secure.
- 2) The IRS control link from the gateway to the IRS controller is assumed to be secure.

2) *RSSI Measurement*: The gateway is capable of measuring the RSSI of the received signal, and this measurement process is secure.

3) *Locations*: IoT devices register all locations with the server. The locations of authentic IoT devices, $\mathbf{L} = [\mathbf{l}_{r,1}, \mathbf{l}_{r,2}, \dots, \mathbf{l}_{r,n}, \dots, \mathbf{l}_{r,N}] \in \mathbb{R}^{N \times 3}$ are stored on the server, where N is the number of IoT devices, $n \in [1, N]$, and

$\mathbf{l}_{r,n} = [x_{r,n}, y_{r,n}, z_{r,n}] \in \mathbb{R}^{1 \times 3}$ denotes the Cartesian coordinates of the n th location.

4) *IRS Configuration*: The IRS controller is the only component that can configure the IRS to reflect signals emitted from a certain direction into a specified direction. When setting up the IRS, the controller initially acquires the target location from the server through the gateway. Subsequently, the IRS is configured based on this target location.

5) *Transmission Power*: The gateway employs adaptive transmit power to maintain a similar level of received signal power, irrespective of whether an IRS is deployed.

6) *Notations*: In this article, $\cdot$, $\odot$, and $\otimes$ denote the dot product, Hadamard product, and Kronecker product, respectively. $\arg\{*\}$ denotes the phase of its complex entry.

V. PROPOSED LOCATION VERIFICATION

A. Channel Formulation

We consider the IRS to have M reflecting elements, where each reflecting element has an area size of A and is capable of introducing an amplitude change of α_m and a phase shift of $\psi_m \in [0, 2\pi)$. The reflection coefficients of the IRS can be expressed as $\Theta = \text{diag}\{\alpha_1 e^{j\psi_1}, \dots, \alpha_m e^{j\psi_m}, \dots, \alpha_M e^{j\psi_M}\}$. Let $h_d \in \mathbb{C}$, $\mathbf{h}_{ts} \in \mathbb{C}^{1 \times M}$, and $\mathbf{h}_{sr} \in \mathbb{C}^{M \times 1}$ denote the channel gain directed from the gateway to the receiver, the gateway to the IRS, and reflected from the IRS to the receiver, respectively. Therefore, the effective channel gain between the gateway and the receiver is

$$h = h_d + \mathbf{h}_{ts} \Theta \mathbf{h}_{sr}. \quad (3)$$

Based on the system model introduced in Section IV-A, the channel gain between the gateway and the IRS and between the IRS and the IoT device can be represented as

$$\mathbf{h}_{ts} = \boldsymbol{\rho}_{ts} \odot \mathbf{u}_{ts} \quad (4)$$

$$\mathbf{h}_{sr} = \boldsymbol{\rho}_{sr} \odot \mathbf{u}_{sr} \quad (5)$$

where $\boldsymbol{\rho}_{ts} \in \mathbb{R}^{1 \times M}$ and $\boldsymbol{\rho}_{sr} \in \mathbb{R}^{M \times 1}$ account for path losses, and $\mathbf{u}_{ts} \in \mathbb{R}^{1 \times M}$ and $\mathbf{u}_{sr} \in \mathbb{R}^{M \times 1}$ account for Rician fading from the gateway to the IRS and Rician/Rayleigh fading from the IRS to the IoT device, respectively.

As mentioned earlier, the gateway and IoT devices are located in the far-field of the IRS. Then, the path loss and the LoS component of the fading coefficient of the channel between the device and the m th element of the IRS are given by

$$\rho_{ts,m} = \rho_{ts} = \frac{\lambda_c \sqrt{G}}{4\pi d_{ts}} \quad (6)$$

$$\bar{\mathbf{u}}_{ts} = \boldsymbol{\mu}_{ts,x} \otimes \boldsymbol{\mu}_{ts,z} \quad (7)$$

where d_{ts} is the distance from the device to the center of the IRS, while $\boldsymbol{\mu}_{ts,x}$ and $\boldsymbol{\mu}_{ts,z}$ are the horizontal and the vertical array response with spatial frequency ϕ_x^{ts} and ϕ_z^{ts} , respectively.

Similarly, for the channel between the IRS and the gateway

$$\rho_{sr,m} = \rho_{sr} = \frac{\lambda_c \sqrt{G}}{4\pi d_{sr}} \quad (8)$$

$$\bar{\mathbf{u}}_{sr} = \boldsymbol{\mu}_{sr,x} \otimes \boldsymbol{\mu}_{sr,z} \quad (9)$$

where d_{sr} refers to the distance from the center of the IRS to the gateway, while $\boldsymbol{\mu}_{sr,x}$ and $\boldsymbol{\mu}_{sr,z}$ are the horizontal and the vertical array responses with spatial frequencies ϕ_x^{sr} and ϕ_z^{sr} , respectively,

$$\boldsymbol{\mu}_{\iota,v} = [1, e^{j\phi_v}, \dots, e^{j(M_v-1)\phi_v}]^T, \quad \iota \in \{ts, sr\}, v \in \{x, z\} \quad (10)$$

$$\phi_x^{\iota} = 2\pi \frac{d_{\iota}}{\lambda} \cos(\varphi^{\iota,e}) \quad (11)$$

$$\phi_z^{\iota} = 2\pi \frac{d_{\iota}}{\lambda_c} \sin(\varphi^{\iota,e}) \sin(\varphi^{\iota,a}) \quad (12)$$

where $\iota \in \{ts, sr\}$ and $\varphi^{\iota,a}$ and $\varphi^{\iota,e}$ are the azimuth and elevation angle of arrival (AoA) or angle of departure (AoD).

B. Acceptance Range

We further elaborate on the channel gain as the summation of subchannels associated with each reflecting element of the IRS, as demonstrated below:

$$h = h_d + \sum_{m=1}^M \alpha_m e^{j\psi_m} h_{ts,m} h_{sr,m} \quad (13)$$

where $h_{ts,m}$ and $h_{sr,m}$ correspond to the m th reflecting element of the vectors $\mathbf{h}_{ts}$ and $\mathbf{h}_{sr}$, respectively. They signify the channel gains from the gateway to the m th reflecting element of the IRS and from the m th reflecting element of the IRS to the receiver, respectively.

Due to the reciprocity of antenna arrays, the definition above remains consistent when the roles of transmitter and receiver are interchanged. Based on this effective channel gain formula, we can calculate the RSS of the signal sent from the IoT device at the gateway as

$$\text{RSS} = P \left| h_d + \sum_{m=1}^M \alpha_m e^{j\psi_m} h_{ts,m} h_{sr,m} \right|^2. \quad (14)$$

This can be rewritten as

$$\begin{aligned} \text{RSS} &= P \left| |h_d| e^{j\arg\{h_d\}} \right. \\ &\quad \left. + \sum_{m=1}^M \alpha_m e^{j\psi_m} |h_{ts,m} h_{sr,m}| e^{j\arg\{h_{ts,m} h_{sr,m}\}} \right|^2 \end{aligned} \quad (15)$$

$$\begin{aligned} &= P \left| |h_d| e^{j\arg\{h_d\}} \right. \\ &\quad \left. + \sum_{m=1}^M \alpha_m |h_{ts,m} h_{sr,m}| e^{j(\psi_m + \arg\{h_{ts,m} h_{sr,m}\})} \right|^2 \end{aligned} \quad (16)$$

where $|h_d|$ and $|h_{ts,m} h_{sr,m}|$ denote the magnitudes and $\arg\{h_d\}$ and $\arg\{h_{ts,m} h_{sr,m}\}$ denote the corresponding phases of h_d and $h_{ts,m} h_{sr,m}$, respectively. Hence, under the assumption of identical reflection amplitude across all elements, the optimal phase shift for maximizing the RSS is

$$\psi_m^{\text{opt}} = \arg\{h_d\} - \arg\{h_{ts,m} h_{sr,m}\} \quad \forall m \in M. \quad (17)$$

We consider the scenario where the IRS is ideally configured to reflect signals from the IoT device to the gateway, that is, the phase shift of each reflecting element is set to an

optimal phase shift ψ_m^{opt} to maximize the RSS. Based on this configuration, the RSS of signal transmitters at other positions in 3-D space will be

$$\hat{\text{RSS}} = P \left| h_d + \sum_{m=1}^M e^{j\psi_m^{\text{opt}}} h_{ts,m} h_{sr,m} \right|^2 \quad (18)$$

where h_d denotes the direct channel gain between the gateway and a specific point, while $h_{sr,m}$ denotes the channel gain between the IRS's m th element and that point.

The legitimacy of an IoT device is determined by evaluating whether its RSS falls within the specified threshold range, denoted as $[\gamma_l, \gamma_r]$. If $\text{RSS} \in [\gamma_l, \gamma_r]$, the device is verified as legitimate; otherwise, if $\text{RSS} \notin [\gamma_l, \gamma_r]$, the authentication is rejected. Similarly, signal transmitters located at other positions should meet the acceptable RSS criteria

$$\hat{\text{RSS}} \in [\gamma_l, \gamma_r]. \quad (19)$$

As discussed in Section V-A, the channel gains $h_{ts,m}$, $h_{sr,m}$, and $h_{sr,m}$ are influenced by the location of the device $\mathbf{l}_t$, the IRS $\mathbf{l}_s$, and the gateway $\mathbf{l}_r$. Therefore, the acceptable range of legitimate device locations can be determined using these.

C. Effectiveness of the IRS

To evaluate the effectiveness of IoT devices equipped with the IRS, we studied a scenario where the number of IRS reflective elements is $M = 121$, in which the gateway, IRS center, and IoT devices are, respectively, located at $\mathbf{l}_r = [20, -37, -8]$, $\mathbf{l}_{sc} = [0, 0, 0]$, and $\mathbf{l}_t = [53, 76, -3]$. The signal is transmitted at frequency $f_c = 47$ GHz. IRS–gateway channel is a Rician fading channel with Rician factor $\kappa = 7$. Device–IRS and device–gateway channels are Rayleigh fading channels. To observe the impact of IRS, we defined the RSS ratio as $\text{RSS}/\text{RSS}_d - 1$, where RSS is the received signal strength and RSS_d is the RSS from the direct path.

Under the premise that the IRS is ideally configured toward the device's registered location, we gradually deviate the device from its registered location in the xy plane. The coordinate information of the gateway, IRS, and device is presented in the upper half of Fig. 3(a) for reference. The RSS ratio on the gateway side, varying with location offsets, is illustrated in Fig. 3(b). The results indicate that when the device is at its registered location $[53, 76, -3]$ (deviation $[0, 0]$), the RSS ratio increases from 0 to 1.90 after deploying an IRS. Additionally, it is noteworthy that the RSS ratio surface displays two peaks. One peak (26.53) is evident when the device deviates to $[-33, -113]$, and another peak (62.71) is observed when the device deviates to $[-53, -76]$. By juxtaposing this with the coordinate information, it can be observed that these two peaks correspond, respectively, to the positions of the gateway and the IRS. The scenario without IRS is shown in Fig. 3(c). We can find that there is only one peak (22.57) at the deviation of $[-33, -113]$, indicating that the maximum RSS ratio is achieved when the device moves to the location of the gateway. Comparing Fig. 3(b) and (c), we observe that the use of the IRS effectively reflects the signal, leading to a significant enhancement in the RSS.

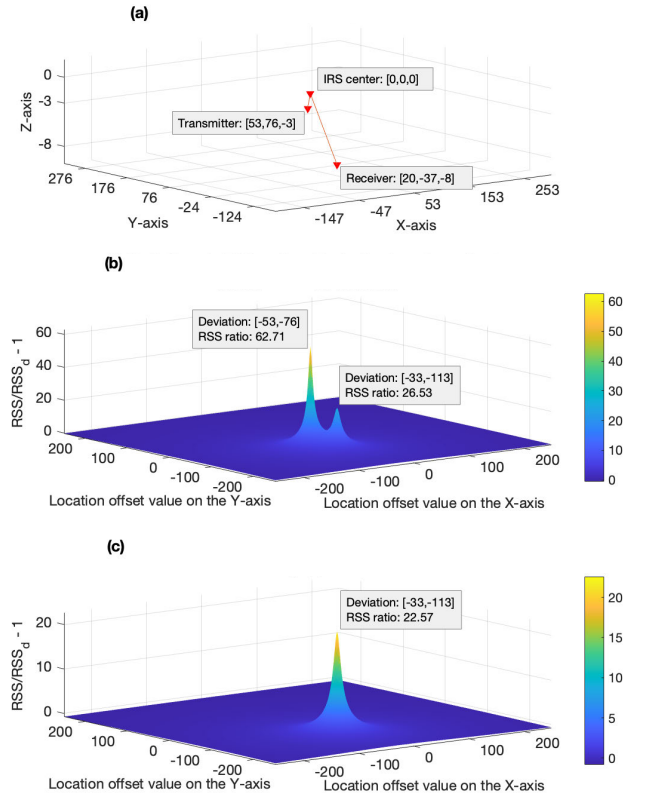


Fig. 3. (a) Coordinates of the transmitter, IRS, and receiver. (b) Variations in the RSS ratio with device location offsets when equipped with the IRS and (c) not equipped with the IRS.

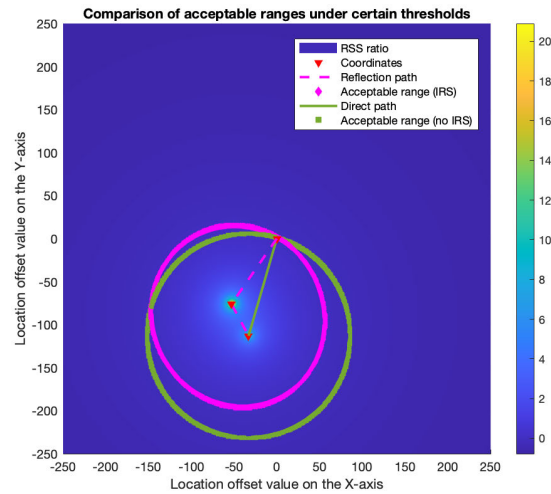


Fig. 4. Comparison of acceptable ranges with or without the IRS under certain thresholds ($\gamma = 0.005$).

We define the range of deviations that can cause the RSS ratio to be between thresholds ($|\text{RSS ratio}| \leq \gamma$) as the acceptable range. To compare the size of the acceptance range between scenarios with and without the IRS, we normalize the RSS by setting the RSS at the registered location as RSS_d , such that the normalized RSS (RSS_n) is 0 when the deviation is $[0, 0]$. The accepted ranges with and without the IRS when the threshold is 0.005 are plotted in pink and green,

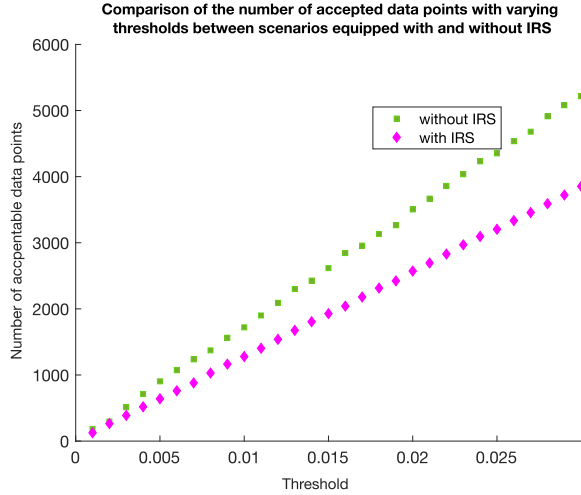


Fig. 5. Comparison of the number of accepted data points with and without the IRS as the threshold varies.

respectively, in Fig. 4. Dashed circles are drawn for reference. The results show that without the IRS, the acceptable range is nearly circular, centered around the gateway location. With the IRS, the acceptable range has its center positioned between the gateway and the IRS, and is obviously not circular. Moreover, the acceptable range with the IRS is significantly smaller than that without the IRS.

To further investigate and compare the size of the acceptable range with and without the IRS, we recorded the number of accepted data points for both scenarios as the threshold varied. The results are presented in Fig. 5. It is evident that the number of accepted data points with the IRS consistently remains lower than that without the IRS, and this discrepancy becomes more pronounced as the threshold increases. This observation validates that the use of the IRS effectively narrows the acceptable range, thereby diminishing the opportunities for potential attackers.

D. Robustness Analysis in Practice

While previous sections establish authentication performance under ideal conditions, practical deployment requires addressing hardware imperfections and environmental variations. This section analyzes system robustness to phase shift errors and environmental interference.

1) *Phase Shift Error Tolerance*: In practical IRS implementations, the ideal phase shift of each reflecting element cannot be perfectly realized due to hardware and environmental imperfections. We model cumulative phase errors from three major sources

$$\Delta\psi_m^{\text{total}} = \Delta\psi_m^q + \Delta\psi_m^c + \Delta\psi_m^e. \quad (20)$$

- 1) *Quantization Error* [50]: Discrete phase shifter implementations introduce quantization following a uniform distribution:

$$\Delta\psi_m^q \sim \mathcal{U}\left[-\frac{\pi}{L}, \frac{\pi}{L}\right] \quad (21)$$

where $L = 2^n$ is the number of quantization levels for n -bit phase shifters.

TABLE II

RSS PERFORMANCE UNDER REALISTIC PHASE ERROR MODELS

Quantization Scheme	Performance Ratio	Degradation
1-bit ($\pm 90^\circ$)	99.75%	$< 0.25\%$
2-bit ($\pm 22.5^\circ$)	99.98%	$< 0.02\%$
3-bit ($\pm 11.25^\circ$)	99.99%	$< 0.01\%$

- 2) *Calibration Error* [51]: Initial channel estimation and phase calculation introduce zero-mean Gaussian measurement errors

$$\Delta\psi_m^c \sim \mathcal{N}(0, \sigma_c^2), \quad \sigma_c = 5^\circ - 10^\circ. \quad (22)$$

This represents the typical accuracy achievable with standard calibration procedures in practical IRS systems.

- 3) *Environmental Drift* [52]: Temperature variations, mechanical vibrations, and humidity affect phase shifter characteristics. We model environmental effects as

$$\Delta\psi_m^e \sim \mathcal{N}(0, \sigma_e^2), \quad \sigma_e = (0.3^\circ - 0.7^\circ) / ^\circ\text{C} \quad (23)$$

corresponding to typical indoor temperature stability ($\pm 1\text{C}$ with $0.5^\circ/\text{C}$ sensitivity).

We conducted Monte Carlo simulations (1000 trials per configuration) using the system configuration from Section III-C. For each trial, phase errors are generated according to (20) with $\sigma_c = 5^\circ$ and $\sigma_e = 0.5^\circ$, applied to the optimal phase shifts (17), and RSS performance is evaluated.

Table II summarizes the results across three commercial quantization schemes: 1-bit, 2-bit, and 3-bit. Results demonstrate exceptional robustness across all commercial implementations. Even 1-bit quantization achieves 99.82% performance, validating practical deployment feasibility. Quantization dominates error variance (86.9%–99.5%), while the large IRS array ($M = 121$) provides strong diversity gain through spatial averaging.

2) *Environmental Interference Robustness*: Our system inherently addresses environmental interference through its Rician channel model (1) and $\kappa = 7$ is configured for effectiveness evaluation. This model captures environmental effects through the 12.5% NLoS component (multipath, shadowing, and interference). The IRS array provides robustness through the following characteristics.

- 1) *Spatial Diversity*: $M = 121$ elements average environmental variations through coherent combining.
- 2) *Geometric Stability*: Location-specific phase patterns are invariant to environmental fluctuations.
- 3) *Enhanced SNR*: The RSS ratio of 1.90 provides 4.6-dB improvement over direct path.

The authentication mechanism's inherent robustness validates deployment feasibility without additional interference mitigation.

VI. PROPOSED AUTHENTICATION PROTOCOL

A. Device Registration

IoT devices register their identity (ID), password, and location with the server through a gateway. The gateway measures

TABLE III
DATA FORMAT STORED BY THE SERVER

Identity	Password	Pseudo-identity	Gateway Identity	Locations	Coarse RSS
ID_i	PW_i	PID_i	GID_1	$l_{i,1}^1$	$CR_{i,1}^1$
			$\dots$	$l_{i,1}^2$	$CR_{i,1}^2$
			GID_w	$l_{i,w}^j$	$CR_{i,w}^j$
$\dots$	$\dots$	$\dots$	$\dots$	$\dots$	$\dots$

¹ i refers to the i -th IoT device,

² w refers to the w -th gateway,

³ j refers to the j -th legitimate location of a device.

the RSS during the registration phase as the coarse RSS. The server stores this information in the format shown in Table III. The IoT device and the server initialize a pseudo-identity PID_i for authentication, where $PID_i^0 = h(ID_i \parallel PW_i)$.

B. Mobility

A device can register at multiple legitimate locations within a gateway's coverage area or across multiple gateways.

The server estimates a deviation $\delta_{i,w}^j$ for each location based on the device's coarse RSS at that location to form an RSS acceptable range $[CR_{i,w}^j - \delta_{i,w}^j, CR_{i,w}^j + \delta_{i,w}^j]$. In the subsequent authentication process, the server identifies the device's specific legitimate location $l_{i,w}^j$ based on the ID_i , GID_w , and the RSS acceptable range.

When an IoT device moves from one registered location to another, it sends its current location to the server through the nearest accessible gateway for reauthentication.

C. Authentication

The specific steps of the proposed authentication protocol are detailed as follows, as illustrated in Fig. 6.

- 1) IoT device i initiates authentication by transmitting the pseudo-identification PID_i in plain text and its location L_i encrypted by $y = h(PID_i \parallel PW_i)$.
- 2) Gateway w receives the signal transmitted by device i and measures the RSS value, $RSS_{i,w,1}$, at time t_1 . It then forward PID_i and $M1$ to the server while appending the RSS information, denoted as $(GID_w, t_1, RSS_{i,w,1})$, related to device i .
- 3) The server retrieves the device's ID_i and PW_i corresponding to the received PID_i . It then computes the shared secret $y = h(PID_i \parallel PW_i)$ and decrypts msg_1 to obtain the device location L_i . Subsequently, the coarse RSS of the device, $CR_{i,w}^j$, is queried based on the ID_i , GID_w , and L_i . The server verifies if $RSS_{i,w,1}$ within the acceptable RSS range $[CR_{i,w}^j - \delta_{i,w}^j, CR_{i,w}^j + \delta_{i,w}^j]$. On the premise of successful verification, the server stores $(GID_w, t_1, RSS_{i,w,1})$ for subsequent authenticate comparison. Then, the server generates a random number N_s and appends it with its pseudo-identity PID_i , location L_i , and the forwarding gateway identity GID_w as $(PID_i \parallel N_s \parallel L_i \parallel GID_w)$. This concatenated data is then encrypted using the shared key PW_i , resulting in msg_2 . The server sends L_i and msg_2 to the gateway.

- 4) Gateway w communicates with the IRS controller to set the reflective link $G \leftrightarrow L_i$ based on L_i . Then, the gateway forward msg_2 to device i .
- 5) If device i receives msg_2 from gateway w , it retrieves N_s by applying the shared key PW_i . Then, it calculates $msg_3 = h(PID_i \parallel PW_i \parallel GID_w \parallel N_s)$ and transmits PID_i and msg_3 to the gateway w . The IoT device generates a new pseudo-identity $PID_i' = h(ID_i \parallel N_s)$ for the future authentication session. The authentication fails if device i does not receive a signal from gateway w before timeout.
- 6) Gateway w measures the RSS value, $RSS_{i,w,1}$, of the received signal at time t_1 . It then forward PID_i and λ to the server while appending the RSS information, denoted as $(GID_w, t_2, RSS_{i,w,2})$, related to device i .
- 7) The server computes $msg_3 = h(PID_i \parallel PW_i \parallel GID_w \parallel N_s)$ and verifies whether it matches the received msg_3 . On the premise of successful verification, the server compares the RSS value of the two measurements with a threshold γ . If $|(RSS_{i,w,2})/(RSS_{i,w,1}) - 1| \leq \gamma$, the server accepts authentication, updates the pseudo-identity of device i to $h(ID_i \parallel N_s)$ and the coarse RSS of device i at location L_i to $\text{mean}(RSS_{i,w,1} + CR_{i,w}^j)$; if not, the server rejects authentication.

After successful authentication, the session key between the IoT device i and the server is generated as $k_{IS} = h(ID_i \parallel PW_i \parallel N_s)$. Subsequently, all communication between them is encrypted and decrypted using this session key.

VII. SECURITY ANALYSIS

In this section, we formally analyze the security of the proposed protocol using BAN logic [53]. Due to space limitations, the language and rules of BAN logic are provided in the appendix for reference.

A. Idealization

First, the proposed protocol is idealized as follows.

$M1: D \rightarrow S : \{PID_i, D \xleftrightarrow{PW_i} S\}_{PW_i}, \{L_i, D \xleftrightarrow{y} S\}_y$ from PID_i .

$M2: S \rightarrow D : \{PID_i \parallel N_s \parallel L_i \parallel GID_w, D \xleftrightarrow{PW_i} S\}_{PW_i}$.

$M3: D \rightarrow S : \{PID_i \parallel GID_w \parallel N_s, D \xleftrightarrow{PW_i} S\}_{PW_i}$ from PID_i .

Here, D and S represent the device and the server, respectively; $y = h(PID_i \parallel PW_i)$ in $M1$ is idealized to $\{PID_i\}_{PW_i}$ and $h(PID_i \parallel PW_i \parallel GID_w \parallel N_s)$ in $M3$ is idealized to $\{PID_i \parallel GID_w \parallel N_s\}_{PW_i}$, because PW_i is a shared key between the device and the server.

B. Initial Assumptions

The following initial assumptions are introduced:

$$D \equiv D \xleftrightarrow{K} S, \quad D \equiv D \xleftrightarrow{Y} S, \quad D \equiv S \Rightarrow N_s$$

$$D \equiv \#(L_i)$$

$$S \equiv D \xleftrightarrow{K} S, \quad S \equiv D \xleftrightarrow{Y} S, \quad S \equiv D \Rightarrow PID_i$$

$$S \equiv \#(PID_i), \quad S \equiv \#(L_i), \quad S \equiv \#(N_s).$$

There is a shared key $K = PW_i$ and a shared secret $Y = h(PID_i \parallel PW_i)$ between the server and the device. Since N_S is generated by the server, the server believes in the freshness of N_S , and the device believes that the server has jurisdiction over N_S . The device believes the freshness of L_i since it was transmitted by itself. Since PID_i was transmitted by the device for initializing authentication, the server believes that the device has jurisdiction over PID_i . Since PID_i and L_i are stored at the server, the server believes in the freshness of PID_i and L_i .

C. Desired Conclusions

1) *Pseudo-Identity*: To initiate an authentication, there must be an agreement on PID_i between the device and the server. Hence, the desired conclusions are as follows.

- 1) $S \models D \models PID_i$: The server believes that the device believes in PID_i .
- 2) $S \models PID_i$: The server believes in PID_i .
- 3) $D \models S \models PID_i$: The device believes that the server believes in PID_i .

2) *Device Location*: To successfully configure the IRS, there must be an agreement on L_i between the device and the server. Hence, the desired conclusions are as follows.

- 1) $S \models D \models L_i$: The server believes that the device believes in L_i .
- 2) $S \models L_i$: The server believes in L_i .

3) *Trusted Nonce*: For authentication to succeed, a trusted N_S is required to update the pseudo-identity and generate session keys. Since N_S is generated by the server, the desired conclusions are as follows.

- 1) $D \models S \models N_S$: The device believes that the server believes in N_S .
- 2) $S \models N_S$: The server believes in N_S .
- 3) $S \models D \models N_S$: The server believes that the device believes in N_S .

D. Security Proof for the Proposed Protocol

1) $M1: S \triangleleft (\{PID_i\}_{PW_i}, \{L_i\}_y)$: As shown in the analysis in Fig. 7, after applying rules $R6$, $R1$, $R3$, and $R4$ to $M1$, we conclude that the server believes that the received pseudo-identity PID_i is from the device i . Therefore, the server believes that $y = h(PID_i \parallel PW_i)$ is the secret shared between the device and the server, represented as $S \models D \stackrel{y}{\rightleftharpoons} S$.

Based on the shared secret y , we can derive the $\{L_i\}_y$ part of $M1$. As shown in Fig. 8, by applying rules $R6$, $R2$, $R3$, and $R4$ to $M1$, we conclude that the server believes the device is located at L_i .

2) $M2: D \triangleleft \{PID_i \parallel N_S \parallel L_i \parallel GID_w\}_{PW_i}$: Fig. 9 shows the step-by-step analysis of $M2$. By applying rules $R1$, $R5$, $R3$, and $R7$ to $M2$, we conclude that the device believes that the server believes in PID_i , N_S , L_i , and GID_w . We can further analyze as follows:

$$\frac{D \models S \models N_S, D \models S \models N_S}{D \models N_S} (R4)$$

which proves that the device believes in N_S .

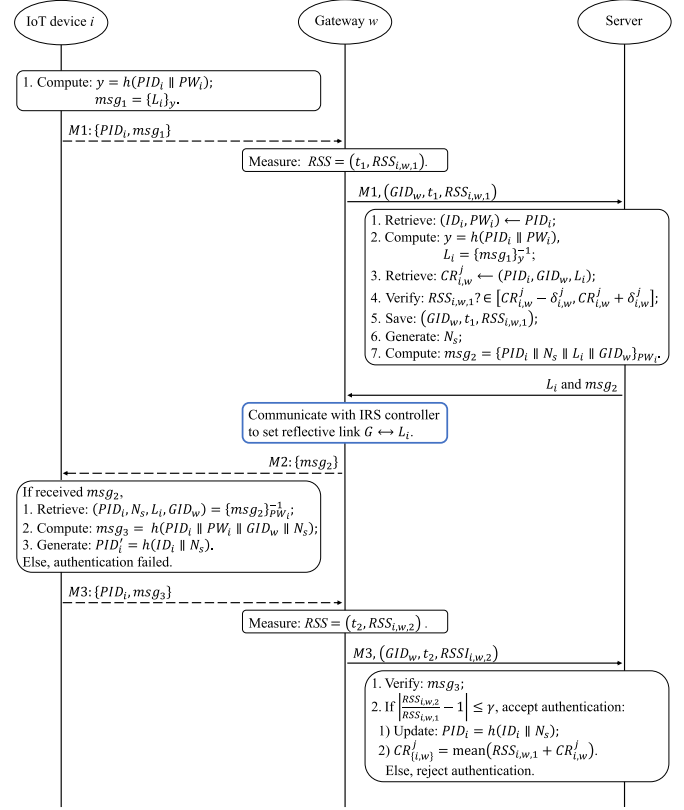


Fig. 6. Authentication protocol.

TABLE IV
REAUTHENTICATION FREQUENCY FOR TYPICAL IOT
MOBILITY SCENARIOS

Scenario	Location Changes /Hour	Dwell Time	f_{loc} (Hz)
Static sensors	0	∞	0
Environment monitoring	1–2	30–60 min	0.0003–0.0006
Warehouse AGVs	6–12	5–10 min	0.002–0.003
Indoor patrol robots	3–6	10–20 min	0.0008–0.002
Smart building devices	2–4	15–30 min	0.0006–0.001

3) $M3: \{PID_i \parallel GID_w \parallel N_S\}_{PW_i}$: The analysis in Fig. 10 shows that, by applying rules $R1$, $R5$, $R3$, and $R7$ to $M3$, we conclude that the server believes that the device believes in N_S and GID_w .

Overall, all desired conclusions ($S \models D \models L_i$, $S \models N_S$, $S \models D \models N_S$, $D \models S \models N_S$, and $D \models N_S$.) are achieved. The proposed protocol has been proved to be secure under BAN logic analysis.

VIII. PERFORMANCE ANALYSIS AND COMPARISON

To discuss the efficiency, effectiveness, and practicality of the proposed authentication mechanism, in this section, we will conduct a comprehensive analysis and comparison with the existing literature in terms of mobility support, security, computational complexity, and communication overhead.

A. Mobility Support Analysis

1) *Reauthentication Frequency Analysis*: The reauthentication frequency in our location-based model depends on the

$$\begin{array}{c}
\frac{S \triangleleft (\{PID_i\}_{PW_i}, \{L_i\}_y)}{S \triangleleft \{PID_i\}_{PW_i}} (R6), S \models D \xleftrightarrow{PW_i} S \\
\frac{S \triangleleft \{PID_i\}_{PW_i}}{S \models D \sim PID_i} (R1), S \models \#(PID_i) \\
\frac{S \models D \sim PID_i}{S \models D \models PID_i} (R3), S \models D \Rightarrow PID_i \\
\hline
S \models PID_i \quad (R4)
\end{array}$$

Fig. 7. Analysis on M1: the server believes the received pseudo-identity.

$$\begin{array}{c}
\frac{S \triangleleft (\{PID_i\}_{PW_i}, \{L_i\}_y)}{S \triangleleft \{L_i\}_y} (R6), S \models D \xleftrightarrow{y} S \\
\frac{S \triangleleft \{L_i\}_y}{S \models D \sim L_i} (R2), S \models \#(L_i) \\
\frac{S \models D \sim L_i}{S \models D \models L_i} (R3), S \models D \Rightarrow L_i \\
\hline
S \models L_i \quad (R4)
\end{array}$$

Fig. 8. Analysis on M1: the server believes the device location.

$$\begin{array}{c}
\frac{D \triangleleft \{PID_i \parallel N_S \parallel L_i \parallel GID_w\}_{PW_i}, D \models D \xleftrightarrow{PW_i} S}{D \models S \sim (PID_i \parallel N_S \parallel L_i \parallel GID_w)} (R1), \frac{D \models \#(L_i)}{D \models \#(PID_i \parallel N_S \parallel L_i \parallel GID_w)} (R5) \\
\frac{D \models S \sim (PID_i \parallel N_S \parallel L_i \parallel GID_w)}{D \models S \models (PID_i \parallel N_S \parallel L_i \parallel GID_w)} (R3) \\
\hline
D \models S \models PID_i, D \models S \models N_S, D \models S \models L_i, D \models S \models GID_w \quad (R7)
\end{array}$$

Fig. 9. Analysis on M2: the device believes that the server believes in PID_i , N_S , L_i , and GID_w .

$$\begin{array}{c}
\frac{S \triangleleft \{PID_i \parallel GID_w \parallel N_S\}_{PW_i}, S \models D \xleftrightarrow{PW_i} S}{S \models D \sim (PID_i \parallel GID_w \parallel N_S)} (R1), \frac{S \models \#(N_S)}{S \models \#(PID_i \parallel GID_w \parallel N_S)} (R5) \\
\frac{S \models D \sim (PID_i \parallel GID_w \parallel N_S)}{S \models D \models (PID_i \parallel GID_w \parallel N_S)} (R3) \\
\hline
S \models D \models N_S, S \models D \models GID_w \quad (R7)
\end{array}$$

Fig. 10. Analysis on M3: the server believes that the device believes N_S and GID_w .

location change frequency f_{loc} , which is application-specific and determined by the operational patterns of IoT devices. Table IV summarizes typical scenarios.

For static or semi-static deployments, location changes are infrequent events.

- 1) *Static Sensors*: No reauthentication required when devices remain at fixed locations.
- 2) *Warehouse Automated Guided Vehicles (AGVs)*: Operating between predefined stations (e.g., loading dock, storage racks, and packaging area), AGVs typically dwell 5–10 min at each location. With 6–12 location changes per hour, $f_{loc} \approx 0.002$ – 0.003 Hz.
- 3) *Indoor Patrol Robots*: Moving between work zones (e.g., office areas, corridors, and conference rooms) with 10–20-min patrol cycles, robots change locations 3–6 times/h, yielding $f_{loc} \approx 0.0008$ – 0.002 Hz.
- 4) *Smart Building Devices*: Mobile environmental sensors or asset trackers moving between rooms or floors, with typical location changes 2–4 times/h, $f_{loc} \approx 0.0006$ – 0.001 Hz.

2) *Authentication Latency and System Overhead*: Each location-triggered reauthentication executes the full authentication protocol (Section VI-C) with estimated latency $T_{auth} \approx 100$ ms [54], [55].

- 1) *Cryptographic Operations (Hash Computations, Pseudo-Identity Generation)*: ~ 30 ms [56].

- 2) *IRS Configuration and RSS Measurement*: ~ 50 ms [57].
- 3) *Message Transmission and Processing*: ~ 20 ms [58].

Since reauthentication is event-driven rather than periodic, the system overhead is determined by

$$\text{Overhead} = f_{loc} \cdot T_{auth}. \quad (24)$$

System throughput, defined as the fraction of time available for data transmission, is

$$\eta = \frac{1}{1 + f_{loc} \cdot T_{auth}}. \quad (25)$$

The overhead and throughput for different scenarios are quantified in Table V, which demonstrates that authentication overhead remains negligible ($<0.03\%$) for typical IoT mobility scenarios, with system throughput exceeding 99.97% in all cases. This minimal overhead is achieved because of the following points.

- 1) Reauthentication is triggered only by actual location changes, not by continuous polling.
- 2) Location changes are infrequent events in fixed/semi-fixed IoT deployments.
- 3) The 100-ms authentication latency is amortized over long dwell times (minutes to hours).

B. Informal Security Analysis

1) *Privacy of the IoT Device*: During the authentication process, a pseudo-identity is used instead of the registered

identity to protect the device's privacy, and this pseudo-identity is updated after each session.

2) *Two-Factor and Mutual Authentication*: In the proposed protocol, the server authenticates the IoT device using its password and location. The server can retrieve the location information for further authentication only if the IoT device uses the correct password and provides a registered location. The protocol employs $h(\text{PID}_i \parallel \text{PW}_i)$ as a shared secret to verify the device's pseudo-identity PID_i , leveraging the fact that PW_i is known exclusively to the device and server. Simultaneously, the server verifies the device's location by checking if the received L_i matches a registered location and if the RSS falls within the acceptable threshold. The device authenticates the server by verifying the retrieved L_i in msg_2 . The correct shared key PW_i is required for the server to generate msg_2 for the device i and derive the shared secret y to decrypt msg_1 and retrieve L_i . The attacker does not have access to PW_i .

3) *Protection Against Impersonation Attack*: Since the identity, password, and location of IoT devices are not transmitted in plain text, attackers cannot obtain this sensitive information. Consequently, they are unable to generate the correct msg_1 to impersonate IoT devices or msg_2 to impersonate servers.

4) *Protection Against Physical Attack*: In the proposed authentication mechanism, the RSS value's compliance with a specified threshold is crucial for accepting or rejecting authentication requests. The use of the IRS increases RSS sensitivity to location changes. Even if an attacker has the credentials of an IoT device, they cannot authenticate from a different location due to the altered RSS values, which would fall outside the acceptable range. If the device is moved from its registered location, the gateway will detect a change in the RSS, leading to authentication failure. This mechanism effectively guards against physical attacks.

5) *Protection Against Replay Attack*: The communication messages msg_1 , msg_2 , and msg_3 are derived from a pseudo-identity that changes with each session. Consequently, messages captured in one session cannot be reused in other sessions, thereby protecting both the device and the server from replay attacks.

6) *Protection Against Man-in-the-Middle (MITM) Attack*: The proposed mechanism uses the IRS to specifically strengthen the signal between the registered IoT device and the gateway. If an attacker pretends to be a registered device at any location and forward the registered device's msg_1 to the server for authentication, the attacker will either not receive the signal from the gateway or will receive only a very weak signal. As a result, the attacker cannot impersonate the server to send messages to the device. Additionally, after configuring the IRS, any attacker's message to the server will be rejected due to an out-of-threshold RSS value. Therefore, the proposed mechanism effectively protects against MITM attacks.

7) *Protection Against DoS Attack*: The proposed protocol is designed to defend against DoS attacks through two mechanisms.

TABLE V
AUTHENTICATION OVERHEAD FOR LOCATION-BASED MOBILITY

Scenario	f_{loc} (Hz)	Overhead (%)	Throughput η (%)
Static sensors	0	0	100.0
Environmental monitoring	0.0005	0.005	99.995
Warehouse AGVs	0.0025	0.025	99.975
Indoor patrol robots	0.0015	0.015	99.985
Smart building devices	0.0008	0.008	99.992

- 1) *Traffic Anomaly Detection*: Messages lacking a valid PID_i are discarded, ensuring only legitimate traffic is processed.
- 2) *Traffic Filtering by the RSS*: Requests with RSS values outside the acceptable threshold are immediately rejected, preventing resource exhaustion from malicious requests.

C. Comparison of Security Threat Resilience

The security threat resistance of the proposed protocol is compared with existing location proximity-based authentication protocols in Table VI. Our analysis shows that the proposed protocol is secure against all major security threats. However, existing works each have security threats that they cannot protect against. Most compared authentication protocols fail to protect device anonymity. Abdurrahman et al. [32] and Chen et al. [35] sent device identities in plain text, fully exposing user privacy. Additionally, [37] fails to protect device anonymity even with private key encryption, as the public key sent alongside the encrypted packet allows attackers to easily obtain the device identity. In [32], the secure transmission of GPS information between devices ensures protection against impersonation, cloning, replay, and MITM attacks, and the manipulation of GPS information is only possible by gaining control of the phone through a physical attack. Additionally, since authentication is based on a hash of a preshared number, GPS location, and timestamp, it is vulnerable to GPS spoofing. Other protocols that rely on CSI [35] or RSS [37] do not use GPS information for location proximity and are thus resistant to GPS spoofing attacks. CSI and RSS are physical-layer characteristics of the wireless channel between the transmitter and the receiver, used as an out-of-band feature to verify a device's location. Due to their high location specificity and difficulty in replicating, the CSI and RSS effectively counter impersonation, replay, and MITM attacks. Zhang et al.'s [37] work is vulnerable to physical and cloning attacks because the session key is transmitted from the mobile phone to the IoT device. Although the session key is encrypted with the IoT device's public key, an attacker who has access to the IoT device's private key can still obtain the session key. Studies by Aman et al. [8], Chen et al. [35], Zhang et al. [37], and Shawky et al. [40] are vulnerable to DoS attacks. In [8], the server initiates PUF verification before RSS verification; the attacker can overwhelm the server by sending numerous false requests. Chen et al. [35] required CSI information from both IoT devices and their related devices; attackers can disrupt the authentication by flooding related devices,

TABLE VI
PERFORMANCE COMPARISON WITH EXISTING LOCATION PROXIMITY AUTHENTICATION PROTOCOL

Security Requirements	Aman et al. [8]	Abdurrahman et al. [32]	Chen et al. [35]	Zhang et al. [37]	Shawky et al. [40]	Proposed Protocol
Device Anonymity	✓	✗	✗	✗	✓	✓
Physical Attack	✓	✗	✓	✗	✓	✓
Impersonation Attack	✓	✓	✓	✓	✓	✓
Cloning Attack	✓	✓	✓	✗	✓	✓
Replay Attack	✓	✓	✓	✓	✓	✓
MITM Attack	✓	✓	✓	✓	✓	✓
DoS Attack	✗	✓	✗	✗	✗	✓
GPS Spoofing Attack	✓	✗	✓	✓	✓	✓
Deployed Security Algorithm	PUF & RSS & Symmetric Crypto	GPS & Symmetric Crypto	CSI & ECC & Blockchain	RSS & Asymmetric Crypto	IRS & RSS & ECDSA	IRS & RSS & Symmetric Crypto

TABLE VII
COMPARISON WITH TRADITIONAL IoT AUTHENTICATION METHODS

Methods	Security Coverage	Computation (Device)	Storage (Device)	Deployment Burden	Mobility Support	Scalability	Key Limitation	Key Strength
Password/PSK	Low-Mod	Very Low	Very Low	Minimal	Weak	High	No physical/location binding	Easy deployment
Certificate/PKI	High	Mod-High	Moderate	High	Good	Moderate	Heavy management overhead	Strong crypto guarantees
Biometric [7]	Mod-High	Moderate	Moderate	Moderate	User-centric	Moderate	Not device/location-oriented	User-specific binding
PUF-based [2]–[4]	High	Low-Mod	Low	Moderate	Neutral	Moderate	Environmental sensitivity	Device-intrinsic binding
ML-based [10]–[15]	Mod-High	High	High	High	Moderate	Low	Resource-intensive	Adaptive learning
Blockchain-based [17]–[22]	High	Mod-High	High	High	Good	Low-Mod	Energy/latency overhead	Decentralized trust
Location/RSSI (plain) [8], [32], [37]	Moderate	Very Low	Very Low	Minimal	Moderate	High	Vulnerable to spoofing/multipath	Lightweight
IRS-aided(Proposed)	High	Low-Mod	Low	Mod-High	Good	High	IRS deployment/-calibration	Enhanced spatial selectivity

Note: Security coverage includes resistance to impersonation, replay, cloning, GPS spoofing, MITM, and DoS attacks (detailed in Table VI).

preventing them from providing sufficient aggregate signatures. Zhang et al. [37] required the smartphone to connect with the IoT device before RSS verification; attackers can flood the smartphone with pairing requests, preventing it from serving the IoT device. In [40], the RSS detection is in the second authentication phase; attackers can compel either the vehicle or the roadside unit to engage in the computationally intensive signature verification processes by retransmitting authentic messages with a falsified timestamp.

D. Comparison With Traditional IoT Authentication Mechanisms

As discussed in Section I, IoT authentication has evolved from password-based schemes to sophisticated approaches incorporating PUFs, biometrics, ML, and blockchain [2], [3], [4], [7], [8], [10], [11], [12], [13], [14], [15], [17], [18], [19], [20], [21], [22], [32], [37]. Each family makes distinct trade-offs across security, efficiency, and deployment complexity. This section positions our IRS-aided scheme relative to these traditional mechanisms. The key tradeoffs are summarized in Table VII.

Password/PSK-based schemes are lightweight but vulnerable to credential theft and lack physical/location binding.

Certificate/PKI-based schemes provide strong cryptographic guarantees but impose heavy provisioning overhead and computational burden unsuitable for constrained devices. *Biometric schemes* [7] offer user-specific binding but require specialized sensors, raise privacy concerns, and do not naturally support autonomous device authentication. *PUF-based schemes* [2], [3], [4] provide unclonable device identity but suffer from environmental sensitivity and do not address location verification. *ML-based* [10], [11], [12], [13], [14], [15] and *blockchain-based schemes* [17], [18], [19], [20], [21], [22] offer adaptive security and decentralized trust, respectively, but introduce significant computational, storage, and energy overhead. *Plain RSSI/location schemes* [8], [32], [37] are lightweight but brittle to environmental variations and spoofing attacks.

Our *IRS-aided approach* strengthens location authentication through controllable radio-environment shaping, achieving enhanced spatial selectivity and robustness while maintaining low device-side computational requirements. Unlike PKI or blockchain, we avoid heavy cryptographic infrastructure; unlike PUF, we leverage the controllable radio environment for better mobility support; unlike plain RSSI, we achieve precise acceptance regions through IRS signal enhancement (Figs. 3 and 4). The tradeoff is IRS infrastructure

TABLE VIII
COMPARISON OF COMPUTATIONAL COMPLEXITY WITH EXISTING PUFs-BASED PROTOCOLS

Protocol	IoT Device	Server
Aman et al. [8]	$N_H + 2N_{MAC} + 2N_{ENC}$	$N_H + 2N_{MAC} + 2N_{ENC}$
Gope et al. [9]	$7N_H + 4N_{ENC}$	$7N_H + 3N_{ENC}$
Shawky et al. [40]	$2N_G + 2N_S + 3N_H + N_{ENC}$	$N_G + 2N_S + 3N_H + N_{ENC}$
Proposed	$3N_H + 2N_{ENC}$	$3N_H + 2N_{ENC}$

TABLE IX
COMPARISON OF COMMUNICATION OVERHEAD WITH
EXISTING PROTOCOLS

Protocol	No. of Messages	No. of bits
Aman et al. [8]	3	824
Gope et al. [9]	3	944
Shawky et al. [40]	3	3264
Proposed	3	808

deployment, which can be amortized across many devices.

E. Comparison of Computational Complexity

The computational complexity of the proposed protocol is compared with [8], [9], and [40] in Table VIII, where N_{MAC} represents the number of message authentication code (MAC) calculation, N_{ENC} represents the number of encryption/decryption operations, N_H represents the number of hash operations, N_G represents the number of ECDSA public key calculations, and N_S represents the number of ECDSA message signing operations. Our proposed protocol and the protocol in [9] both employ hashing and encryption/decryption. In contrast, the protocol in [8] additionally utilizes the MAC technique, while the protocol in [40] is based on ECDSA. We consider the hashing operations, encryption/decryption operations, and the message signing operations in ECDSA to have a time complexity of $\mathcal{O}(n)$, and the public key calculations in ECDSA to have a time complexity of $\mathcal{O}(m^2n)$, where n is the length of the input data and m is the order of the selected base point on the elliptic curve. The MAC operation in [8], based on universal hashing or block ciphers, also has a time complexity of $\mathcal{O}(n)$. Therefore, the time complexity of [8], [9], and [40] and the proposed protocol are $\mathcal{O}(n)$, $\mathcal{O}(n)$, $\mathcal{O}(m^2n)$, and $\mathcal{O}(n)$, respectively. The time complexity of [9] and [40] is significantly greater than that of [8] and our proposed protocol. Although the number of operations in [8] is comparable to our proposed protocol, it involves more MAC operations, which typically consume more time than hashing. Therefore, the proposed protocol has lower computational complexity compared to the existing state of the art.

F. Comparison of Communication Overhead

We adopt the parameters from [8]: identities are 8 bits, nonces for verifications are 48 bits, nonces for session key generation are 128 bits, PUF input challenges and output responses are 128 bits, the MAC operation is 32-bit UMAC,

and the encryption operation uses a 128-bit block cipher. In our proposed protocol, location coordinates are represented with 32 bits to achieve centimeter-level accuracy [59]. The parameters for ECDSA are adopted from [40]. The communication load between the device and the gateway/server is listed in Table IX, which shows that the proposed protocol has lower communication overhead compared with [8], [9], and [40].

IX. APPLICABILITY ANALYSIS AND SCENARIO POSITIONING

This section analyzes the scenarios in which IRS-aided authentication is most applicable and contrasts them with those better served by specialized alternatives to clarify the scope of our contribution. A comprehensive suitability analysis is presented in Table X. The key insight is that different IoT contexts demand different authentication paradigms. Our contribution targets the economically significant niche of infrastructure-rich, fixed/semi-fixed, location-sensitive deployments where IRS-enhanced spatial selectivity provides security advantages unattainable with plain RSSI while maintaining efficiency over PKI/blockchain.

Our scheme achieves suitability in infrastructure-dense, location-sensitive scenarios with fixed or semi-fixed IoT devices.

- 1) Smart buildings and campuses, where indoor IRS deployment is straightforward, device positions are stable (sensors, access control, and asset tracking), and high device density amortizes infrastructure costs.
- 2) Industrial IoT and manufacturing, where location-binding prevents remote control attacks (attackers must be physically present at authorized stations), environmental robustness handles industrial RF interference, and moderate AGV mobility (<20 km/h) falls within tolerance.
- 3) Smart retail, where preventing device relocation attacks (e.g., POS terminal repositioning for skimming) and multistore standardized deployment provide security and cost benefits.
- 4) Healthcare IoT, where stationary or semi-fixed medical devices (e.g., bedside monitors and infusion pumps) benefit from location-bound authentication, ensuring patient safety, preventing unauthorized device access, and supporting compliance with privacy regulations.
- 5) Smart city, where the IRS can enhance physical-layer security for fixed infrastructure such as street sensors, traffic lights, and surveillance cameras, enabling secure location-sensitive services while maintaining low operational overhead across dense urban deployments.

TABLE X
IRS-AIDED AUTHENTICATION: SCENARIO SUITABILITY ANALYSIS

IoT Scenario	Suitability	Key Factors	Advantages	Limitations
Smart buildings & campuses	Highly-5	Indoor IRS deploy; fixed devices; dense.	Room-level spatial precision; low device overhead; tamper detection.	Minor: initial IRS installation.
Industrial IoT & manufacturing	Highly-5	Fixed equipment; security-critical; moderate AGV mobility.	Prevents remote attacks (location-binding); robust to RF interference real-time (<100 ms); AGV support.	Weekly recalibration in variable environments.
Smart retail & commercial	Highly-4	Indoor; fixed POS/sensors; chain economies.	Anti-tampering (POS relocation detection); multi-store shared infra.; zone-based access.	Initial deployment cost for chains.
Healthcare IoT	Moderately-4	Fixed medical equipment; zone-critical.	Medical device location verification; prevents equipment tampering.	Location privacy.
Smart city (localized)	Moderately-3	Fixed sensors at key intersections	Sensor location verification; prevents replacement attacks.	Outdoor cost; large-scale coverage is challenging.
Cross-domain roaming	Limited-2	Multi-org; inter-carrier.	Limited (can integrate as intra-domain layer).	Single-domain architecture; no federated identity; IRS is org-specific.
Vehicular networks (V2X)	Limited-1	High-speed; privacy-sensitive; roadway.	Not applicable	Mobility exceeds tolerance; IRS coverage infeasible; location-binding contradicts anonymity.
Mobile crowdsensing	Limited-1	User mobility; wide-area; anonymity required.	Not applicable	Location privacy; unpredictable positions; IRS cannot cover arbitrary locations.

Note: Suitability ratings (Very High = 5, Unsuitable = 1) indicate the degree of alignment with the strengths of the IRS-aided scheme.

Conversely, our scheme has limited suitability for the following points.

- 1) Cross-domain roaming, where multiorganizational trust requires federated protocols beyond our single-domain architecture (though integration as an intradomain layer with cross-domain MEC protocols is possible).
- 2) Vehicular networks, where high-speed mobility (>60 km/h) exceeds our tolerance, roadway IRS coverage is infeasible, and critically, location-binding contradicts vehicular privacy requirements.
- 3) Mobile crowdsensing, where user mobility is unpredictable and location disclosure conflicts with privacy preservation.

X. CONCLUSION

This article presented a comprehensive IRS-aided authentication mechanism for IoT devices, demonstrating the viability of leveraging IRS technology for robust location-based authentication in resource-constrained environments. Our primary contributions include: 1) integration of advanced IRS technology into IoT authentication systems, enabling enhanced location-based identity verification through controllable signal enhancement; 2) comprehensive robustness analysis demonstrating exceptional performance (99.82%–99.99%) across commercial IRS implementations despite hardware imperfections; 3) mobility support framework achieving minimal system overhead (<0.03%) with location-based reauthentication; 4) explicit scenario positioning analysis showing high suitability for infrastructure-dense deployments while acknowledging limitations for high-mobility applications; and 5) integration of pseudo-identity verification and RSS filtering to enhance DoS attack resilience. The proposed mechanism establishes a foundation for practical, secure, and efficient

IoT authentication in infrastructure-rich deployment scenarios, while future research directions include integration with emerging 6G IRS technologies and adaptive configurations for dynamic IoT environments.

APPENDIX

This section briefly introduces the logic language and rules of BAN logic we use for protocol analysis, following [53], where P and Q represent principals, X and Y represent statements, and K denotes the shared key.

A. Logic Language

The proofs in BAN logic are interpreted as follows:

- 1) $P \models X$: P believes that X is true.
- 2) $P \triangleleft X$: P sees X .
- 3) $P \sim X$: P once said X .
- 4) $P \Rightarrow X$: P has jurisdiction over X .
- 5) $\#(X)$: The formula X is fresh.
- 6) $P \stackrel{K}{\leftrightarrow} Q$: P and Q may use the shared key K to communicate.
- 7) $P \stackrel{K}{\rightleftharpoons} Q$: The formula X is a secret known only to P and Q and possibly to principals trusted by them.
- 8) $\{X\}_K$: The formula X is encrypted under the key K .

B. Rules

For inference, the following rules are defined in BAN logic:

- 1) RI : Message-meaning rules for shared keys

$$\frac{P \models P \stackrel{K}{\leftrightarrow} Q, P \triangleleft \{X\}_K}{P \models Q \sim X}.$$

That is, if P believes that the key K is shared with Q and sees X encrypted under K , then P believes that Q once said X .

2) R2: Message-meaning rules for shared secrets

$$\frac{P \models P \stackrel{Y}{\Leftarrow} Q, P \triangleleft \{X\}_Y}{P \models Q \vdash X}.$$

That is, if P believes that the secret Y is shared with Q and sees X encrypted under Y , then P believes that Q once said X .

3) R3: Nonce-verification rule

$$\frac{P \models \#(X), P \models Q \vdash X}{P \models Q \models X}.$$

That is, if P believes that X is fresh and Q once said X , then P believes that Q believes X .

4) R4: Jurisdiction rule

$$\frac{P \models Q \Rightarrow X, P \models Q \models X}{P \models X}.$$

That is, if P believes that Q has jurisdiction over X then P trusts Q on the truth of X .

5) R5: Freshness rule

$$\frac{P \models \#(X)}{P \models \#(X, Y)}.$$

That is, if one part of the formula is fresh, then the entire formula must also be fresh.

6) R6: Seeing rule

$$\frac{P \triangleleft (X, Y)}{P \triangleleft X}.$$

That is, if a principal sees a formula, then he also sees its components.

7) R7: Belief rule

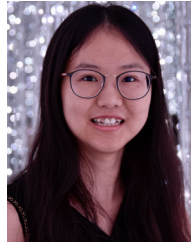
$$\frac{P \models Q \models (X, Y)}{P \models Q \models X}.$$

That is, if a principal believes in a formula, then he also believes in its components.

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